Software Project Management
Assignment-4
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

How many hours is a person month?

a. 122
b. 132
c. 142
d. 152
e. 162
f. 172
g. 720

Correct Answer: d. 152

Detailed Solution:

A person-month is a unit of effort representing the work done by one person in one month. COCOMO II and many project management standards assume 152 hours per person-month (based on 19 working days × 8 hours/day).

QUESTION 2:

Which of the following software effort estimation methods is most suitable when an existing software system is being modified or enhanced?

a. Parametric Models
b. Expert Judgment Technique
c. Analogy-Based Method
d. Price-to-Win Method
e. Similarity-Based Method

Correct Answer: c. Analogy-Based Method

Detailed Solution:

When modifying or enhancing an existing system, the most suitable estimation method is one that compares the proposed changes to similar modifications made previously in the same system or in comparable projects. The Analogy-Based Method does exactly that — it uses historical data from past, similar work to estimate effort for the new changes.

QUESTION 3:

Which one of the following is a key advantage of the Delphi estimation technique compared to estimation by a single expert?

a. It is largely free from the bias of any individual expert.
b. It always produces a quantitative estimate.
c. It explicitly accounts for project-specific peculiarities.
d. It considers that different system components may have different complexities.
e. It requires significantly less time and effort than expert estimation.

Correct Answer: a. It is largely free from the bias of any individual expert

Detailed Solution:

The Delphi estimation technique (including Wideband Delphi) involves multiple experts providing anonymous estimates in iterative rounds, with feedback and revision between rounds. Its key advantage over a single expert is that it reduces individual bias, groupthink, and the undue influence of dominant personalities, leading to a more balanced and reliable estimate.

QUESTION 4:

Which one of the followings is NOT a limitation of algorithmic estimation models?

a. They lack effective mechanisms to account for exceptional situations, such as outstanding teamwork or an unusually good match between skills and tasks.
b. They can be calibrated using data from past projects to improve estimation accuracy.
c. Inaccurate size estimates or cost-driver ratings can lead to unreliable effort estimates.
d. Some important factors, such as experience and team cohesion, are difficult to quantify accurately.
e. They may not adequately capture qualitative aspects of the development environment.

Correct Answer: b. They can be calibrated using data from past projects to improve estimation accuracy.

Detailed Solution:

It is actually an advantage or feature of algorithmic models. Calibration improves accuracy, which is a strength, not a drawback.

QUESTION 5:

Consider a transaction project of 90,000 lines of code. Productivity is about 500 SLOC/person-month. What will be the effort required to develop the project?

a. 60 person-month
b. 90 person-month
c. 120 person-month
d. 180 person-month
e. 240 person-month
f. 300 person-month
g. 500 person-month

Correct Answer: d. 180 person-month

Detailed Solution:

Effort = Size / Productivity

Size = 90,000 SLOC

Productivity = 500 SLOC/person-month

Effort=90,000500=180 person-monthsEffort=50090,000 =180 person-months

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QUESTION 6:

Which one of the followings is NOT true regarding function points?

a. independent of the language, tools, or methodologies used for implementation
b. based on the user's external view of the system
c. measure functionality that the user requests and receives
d. can be estimated early in analysis and design
e. computed from the code implementing the requirements

Correct Answer: e. computed from the code implementing the requirements

Detailed Solution:

Function points are a size metric based on the functional requirements of the system as seen from the user's perspective. They are intentionally designed to be independent of implementation details such as programming language, tools, or actual code.

QUESTION 7:

Which one of the followings is NOT a key component in function point analysis?

a. Number of Inputs
b. Number of Outputs
c. Number of Inquiries
d. Number of Interfaces
e. Number of Operations

Correct Answer: e. Number of Operations

Detailed Solution:

Number of Operations is not a standard component in function point analysis. "Operations" is more relevant to object-oriented metrics (e.g., number of methods) or UML, not to function points.

QUESTION 8:

In a software project, if the degree of influence (DI) for each of the 14 General System Characteristics is 5, then what will be value of VAF (Value Adjustment Factor), using Function Point analysis?

a. 1.07
b. 1.21
c. 2.07
d. 2.21
e. 1.35
f. 70

Correct Answer: e. 1.35

Detailed Solution:

VAF=0.65+70/100 =0.65+0.70=1.35
where, total degree influence, TDI=14×5=70

QUESTION 9:

Which of the followings are limitations of function point analysis?

a. It does not consider algorithmic complexity of a function
b. It is hard to fudge
c. It is language independent
d. It is understandable by clients
e. It requires extensive training

Correct Answer: a. It does not consider algorithmic complexity of a function
e. It requires extensive training

Detailed Solution:

Function point analysis does not account for internal algorithmic complexity, making it less accurate for highly complex functions, and it demands significant training to apply consistently. In contrast, being language-independent, hard to manipulate, and understandable to clients are considered strengths, not limitations.

QUESTION 10:

Consider a project with the following functional units: 80 user inputs, 60 user outputs, 50 user enquiries, 10 user files, 7 external interfaces. What will be the unadjusted function points for the project?

a. 769
b. 869
c. 969
d. 1069
e. 1169

Correct Answer: c. 969

Detailed Solution:

Using standard IFPUG average complexity weights:

External Inputs (EI): 80 × 4 = 320

External Outputs (EO): 60 × 5 = 300

External Inquiries (EQ): 50 × 4 = 200

Internal Logical Files (ILF): 10 × 10 = 100

External Interface Files (EIF): 7 × 7 = 49

Total UFP = 320 + 300 + 200 + 100 + 49 = 969

******THE END******